Pixels, Frames & Transformers

Vision Transformers · Module 08

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An Image Is Worth 16 × 16 Words

In 2020, Dosovitskiy and colleagues took the transformer from language models, changed almost nothing, and fed it images cut into small squares.

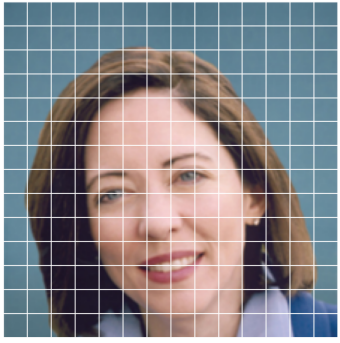
- Each 16 × 16 pixel **patch** becomes a token, like a word
- The same self-attention layers as BERT
- Trained on enough images, it matched the best CNNs

Why it matters for this course
ViTs are inside almost everything that follows: CLIP (module 09), vision-language models (10, 11), open-vocabulary detectors and SAM (12), and the video models of Part 4.

Dosovitskiy et al. (2021), ICLR.

Patches as Tokens

224 × 224 pixels, 14 × 14 patches



rows 7 and 8 as a sequence: patches 85–112 of 196



Each patch is $16 \times 16 \times 3 = 768$ numbers. One linear layer turns it into the patch's token embedding; a learned **position embedding** is added so the model knows where the patch was. A **class token** at the front collects a summary of the whole image.

Official portrait of Sen. Maria Cantwell, public domain, via unitedstates/images.

BERT and ViT, Side by Side

	BERT (text)	**ViT** (images)
input unit	a word piece	a 16×16 pixel patch
token embedding	looked up in a vocabulary	computed from the pixels
order	position embeddings	position embeddings (2D layout learned)
summary token	`[CLS]`	class token
layers	self-attention + MLP	the same
pre-training	guess masked words	image labels, or self-supervised
the usual embedding	`[CLS]` or mean of tokens	class token

If you know the BERT deck of Bags, Vectors & Transformers, you already know most of a ViT. The new part is how the tokens are made.

Every Patch Sees Every Patch

- A CNN filter sees a small neighborhood; opposite corners of the image only meet after many layers
- In a ViT, **self-attention** lets each token take information from every other token, from the first layer on
- The price: attention compares all pairs, so cost grows with the **square** of the number of tokens

image, patch	tokens	attention
224, 32 (CLIP B/32)	50	2,500
224, 16	197	38,809
384, 16	577	332,929
518, 14 (DINOv2)	1,370	1,876,900

Double the image size, and attention costs about 16 times as much. Small details, like text on a sign, can vanish inside a large patch.

Why ViTs Need More Data

A CNN has assumptions built in: nearby pixels belong together, and a pattern means the same thing anywhere in the image. A ViT has to **learn** these from data.

- Trained on ImageNet's 1.3 million images alone, the original ViT did worse than ResNets
- Pre-trained on 14 million images (ImageNet-21k) it caught up, and on 300 million it did better

What this means for you
Never train a ViT from scratch on social science data. Start from a pre-trained one: as a frozen embedding with a linear probe, or fine-tuned with a small learning rate.

Later work (such as DeiT, Touvron et al., 2021) made ViTs trainable on ImageNet alone with heavy augmentation, but the advice stands.

Pre-Training Without Labels

Self-distillation (DINO, DINOv2)
A student network sees small crops of an image, a teacher sees large crops, and the student learns to give the same output as the teacher. No labels, and the embeddings describe images in general.

Masked image modeling (MAE)
Hide 75% of the patches and train the model to reconstruct them, like BERT guessing masked words. What it learns about images transfers to other tasks.

For categories that ImageNet doesn't have, such as *protest*, embeddings from self-supervised models are often a better starting point than those of a model trained to sort ImageNet's 1,000 categories.

Oquab et al. (2024); He et al. (2022).

Attention Maps: Attractive, Not an Explanation

The class token's attention over the patches, laid out on the grid, makes a heat map. But:

- 12 heads per layer often disagree; averaging them is a choice
- One layer's map ignores what earlier layers mixed in; **rollout** follows all layers but adds assumptions (Abnar & Zuidema, 2020)
- Large ViTs park information in a few background tokens, which light up for no visible reason (Darcet et al., 2024)

A map that highlights the protest sign doesn't show that the sign *caused* the prediction.

Test it instead: cover the sign and see whether the prediction changes; check the model on counterexamples (signs without protest, protest without signs).

CNN or ViT for Your Project?

What matters more than the architecture

1. What the model was pre-trained on
2. How many labeled images you have
3. How close your images are to web photos

A sensible order

frozen embedding + linear probe → fine-tune if a learning curve says more labels still help → never from scratch

Reading the scoreboard

On 40 test images, a balanced accuracy of 0.80 has a 95% interval of roughly ± 0.13. Small differences between methods are mostly noise.

And every method learned from Commons category labels, which only approximate your codes.

In the Notebook

- Cutting an image into patches by hand
- Inside `vit-base-patch16-224`: tokens, layers, class token, position embeddings
- Attention maps and attention rollout, and their limits
- Fine-tuning a ViT on the protest labels
- A linear probe on frozen ViT embeddings, and your scoreboard so far

part3_transformers/08_vision_transformers/notebooks/
08_vision_transformers_exercises.ipynb · 6 exercises · GPU needed